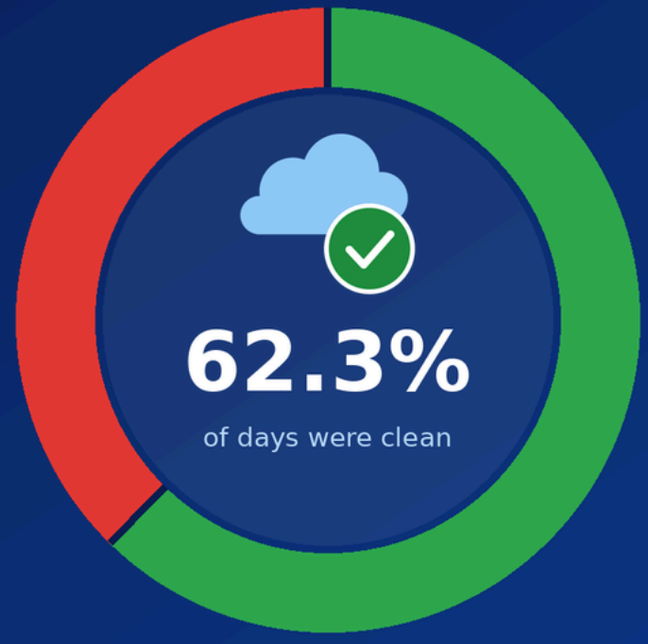
DATA ANALYTICS PROJECT

Air Quality Index (AQI) Analysis Amravati 2024

A full year of air quality data, taken from a raw government source and turned into an interactive dashboard.

366 days of data | Source: CPCB, India | Python + Power BI



Clean: 228 days

Polluted: 138 days

AQI SCALE

Good	Satisfactory	Moderate	Poor	Very Poor
0-50	51-100	101-200	201-300	301-400



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Data Analyst

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85.86 Average AQI

205 Max AQI

16 Min AQI

Swipe to see the dashboard >

Air Quality Index (AQI) Analysis

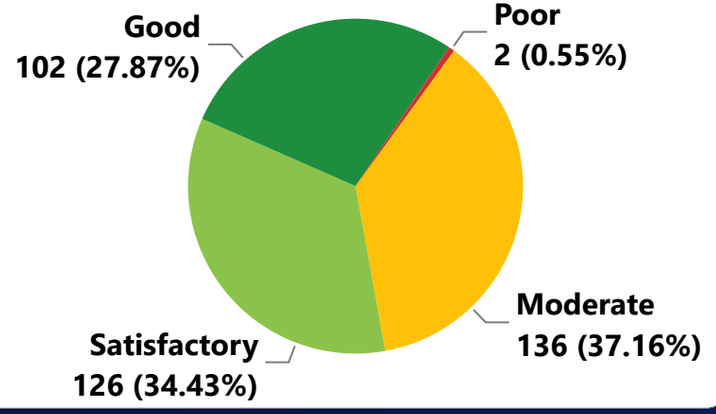
Amravati 2024 | Source - CPCB, India | 366 days of data | Python + Power BI

Home | Dashboard | Conclusion

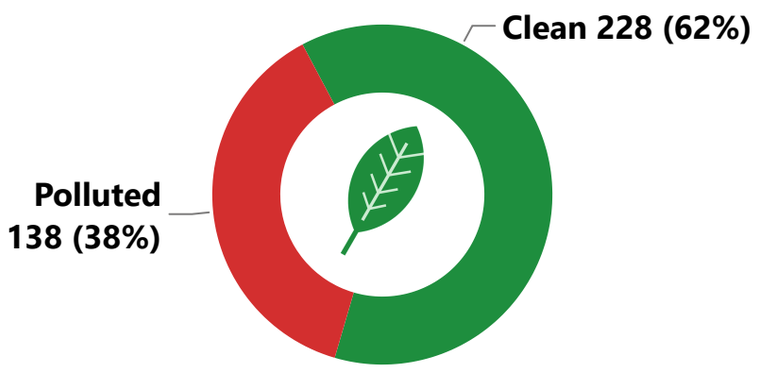
366	85.86	205	16	138	228
Total Days	Average AQI	Max AQI	Min AQI	Polluted Days	Clean Days

January | February | March | April | May | June | July | August | Septem... | October | Novem... | December

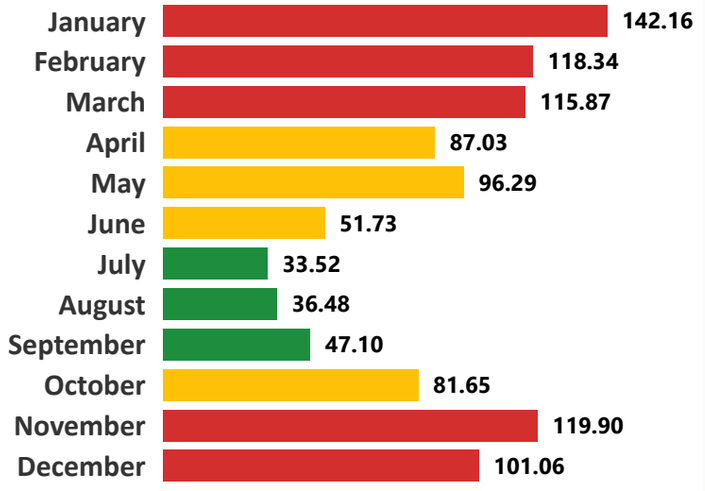
Days by AQI Category



Clean vs Polluted Days



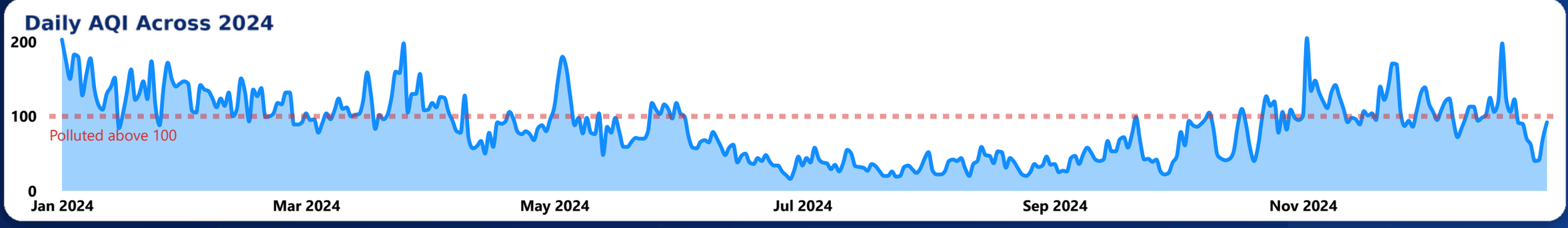
Average AQI by Month



Insight: Nov-Mar peaks, monsoon cleans the air

Conclusion: 62.3% of days were clean

Daily AQI Across 2024



Air Quality Index (AQI) Analysis

Amravati 2024 | Source - CPCB, India | 366 days of data | Python + Power BI

Home | Dashboard | Conclusion

Q1 (Jan-Mar) average AQI
125.62
Above 100

Q2 (Apr-Jun) average AQI
78.55
50 to 100

Q3 (Jul-Sep) average AQI
38.95
Below 50

Q4 (Oct-Dec) average AQI
100.66
Above 100

Key Findings

- 62.3% of days were clean (228 of 366), and 37.7% were polluted (138)
- November to March all averaged above 100 AQI, with January the worst at 142.16
- June to September stayed under 52, with July the cleanest at 33.52
- Only 2 days (0.55%) reached Poor, with a maximum AQI of 205 and a minimum of 16

Recommendations

- Prioritize November to March for any air-quality action
- Extend the pipeline to multiple years to check whether 2024 is typical
- Add a second city to see if the pattern is local or regional
- Add weather and pollution-source data, because this data shows when AQI rises, not why
- Limitation: only one year of data from one city

Bottom line: pollution peaks Nov-Mar and clears in the monsoon, so Nov-Mar is the window to act on.



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Thank you for viewing